

JEE MAIN REVISION MASTER: SOLID STATE OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 67 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Solid State — Standard Calculations & Problem Archetypes	33 PYQs	Quantitative Analysis
Very High	Solid State — Applied Cases, Reactions & Traps	28 PYQs	Applied Topics
High	Solid State — Core Principles & Governing Laws	10 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Solid State — Standard Calculations & Problem Archetypes	$1. \text{ (Nearest integer) [Given : } N_A = 6.022, \quad d = \frac{Z \times M}{a^3 \times N_A}, \quad 10^{21}$
Solid State — Applied Cases, Reactions & Traps	$\rightarrow r_{Cs^{+}} + r_{Cl^{-}} = \frac{\sqrt{3}a}{2}, \quad Z = 2 \text{ for bcc}, \quad \pi r^2$
Solid State — Core Principles & Governing Laws	$\frac{a}{2}, \quad A_4 \times \frac{1}{2} B_4 O_8, \quad \times 10^{23}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Solid State — Standard Calculations & Problem Archetypes (Very High Yield — 33 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Solid State.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "Which of the following expressions is correct in case of a CsCl unit cell (edge length 'a')?..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (Numerical): "Number of crystal systems from the following where body centred unit cell can be found, is _____. Cubic, tetragonal, orthorhombic, hexagonal..." [Ans: 3]
- [2023 Evening Shift] Q#2 (Numerical): "A metal M crystallizes into two lattices :- face centred cubic (fcc) and body centred cubic (bcc) with unit cell edge length of 2.0 and $2\sqrt{2}$..." [Ans: 4]

3.2 Solid State — Applied Cases, Reactions & Traps (Very High Yield — 28 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Solid State.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "A cubic solid is made up of two elements X and Y. Atoms of X are present on every alternate corner and one at the center of cube. Y is at the center of each edge..." [Ans: N/A]
- [2023 Evening Shift] Q#2 (MCQ): "The correct relationships between unit cell edge length 'a' and radius of sphere 'r' for face-centred and body-centred cubic structures..." [Ans: D]
- [2023 Morning Shift] Q#3 (MCQ): "A compound is formed by two elements X and Y. The element Y forms cubic close packed arrangement and those of element X occupy one half of the tetrahedral voids..." [Ans: N/A]

3.3 Solid State — Core Principles & Governing Laws (High Yield — 10 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Solid State.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Solid State.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Solid State PYQs.

Representative Source PYQs:

- [2023 Evening Shift] Q#3 (Numerical): "Iron oxide FeO, crystallises in a cubic lattice with a unit cell edge length of 5.0 Å. If density of the FeO in the crystal is 4.0 g cm^{-3} ..." [Ans: 4]
- [2023 Morning Shift] Q#4 (MCQ): "Which of the following represents the lattice structure of $\text{Al}_2(\text{OH})_4\text{O}_3$ containing Al^{2+} , Al^{3+} and O^{2-} ions?..." [Ans: N/A]
- [2022 Morning Shift] Q#3 (Numerical): "An element M crystallises in a body centred cubic unit cell with a cell edge of 300 pm. The density of the element is 6.0 g cm^{-3} . The atomic mass of M is..." [Ans: 22]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 15th April Morning Shift (Q#1)

Which of the following expressions is correct in case of a CsCl unit cell (edge length 'a')?

- (A) $r_{\text{Cs}^+} + r_{\text{Cl}^-} = \frac{\sqrt{3}}{2}a$
- (B) $r_{\text{Cs}^+} + r_{\text{Cl}^-} = \frac{a\sqrt{2}}{2}$
- (C) $r_{\text{Cs}^+} + r_{\text{Cl}^-} = a$
- (D) $r_{\text{Cs}^+} + r_{\text{Cl}^-} = (a / 2)$

Correct Answer: None

Solution Analysis: CsCl has body centered type structure in which Cs^+ occupies at corner of a cube and Cl^- occupies the centre of the cube. $2r_{\text{Cs}^+} + 2r_{\text{Cl}^-} = \sqrt{3}a$ (where a is the edge length of the cube) $r_{\text{Cs}^+} + r_{\text{Cl}^-} = \frac{\sqrt{3}}{2}a$

Benchmark Problem 2: JEE Main 2023 (Online) 25th January Morning Shift (Q#1)

A cubic solid is made up of two elements X and Y. Atoms of X are present on every alternate corner and one at the center of cube. Y is at $(1/3)^{\text{rd}}$ of the total faces. The empirical formula of the compound is :

- (A) $\text{X}_2\text{Y}_{1.5}$
- (B) X_3Y_2
- (C) $\text{XY}_{2.5}$
- (D) $\text{X}_{2.5}\text{Y}$

Correct Answer: None

Solution Analysis: Number of X particles = $4 \times (1/8) + 1 = 1.5$ Number of Y particles = $6 \times (1/3) \times (1/2) = 1$ $\therefore$ Empirical formula = $\text{X}_{1.5}\text{Y}_1 = \text{X}_3\text{Y}_2$

Benchmark Problem 3: JEE Main 2023 (Online) 30th January Evening Shift (Q#3)

Iron oxide FeO, crystallises in a cubic lattice with a unit cell edge length of 5.0 Å. If density of the FeO in the crystal is 4.0 g cm^{-3} , then the number of FeO units present per unit cell is _____. (Nearest integer) Given: Molar mass of Fe and O is 56 and 16 g mol^{-1} respectively. $N_A = 6.0 \times 10^{23} \text{ mol}^{-1}$

Correct Answer: 4

Solution Analysis: $d = \frac{z \times M}{a^3 \times N_A}$ $4 = \frac{z \times 72}{(5 \times 10^{-8})^3 \times 6 \times 10^{23}}$ $z = \frac{4 \times 72 \times 125 \times 10^{-1}}{726} \approx 4$